

INTERNET PRIVACY BASICS: VPNs, TOR, ONION & PRIVATE BROWSERS

There are several different tools available for increasing privacy while using the internet. A **VPN, Tor, an Onion site, and a privacy-focused browser are not the same thing**. Each provides a different type of protection, and some can be used together.

A simple way to understand the different layers is:

Privacy Browser → Reduces online tracking

VPN → Protects and reroutes your internet connection

Tor → Routes browsing through multiple independent relays

.onion → Websites and services accessed through the Tor network

None of these tools automatically makes someone completely anonymous. What they protect depends on how they are used.

FREE OPTIONS

TOR BROWSER — FREE

Website: torproject.org

Tor Browser is the main browser designed specifically for accessing the **Tor network**.

Instead of connecting directly from your device to a website, Tor sends the connection through multiple relays before it reaches its destination. This makes it significantly more difficult for the destination website to determine your original IP address.

Tor Browser also includes protections designed to reduce browser fingerprinting and tracking.

Tor Browser can access both:

- Regular websites
- **.onion websites**

Available for:

- Windows
- Mac
- Linux
- Android phones and tablets

There is currently **no official Tor Browser for iPhone or iPad**. The Tor Project recommends **Onion Browser** for iOS users, although Apple's technical restrictions mean it cannot provide exactly the same protections as Tor Browser on Android or computers.

Good for: Stronger browsing privacy, concealing your originating IP from websites, censorship resistance, and accessing legitimate .onion services.

ONION BROWSER — FREE

For: iPhone and iPad

Onion Browser is an open-source browser that allows iPhone and iPad users to browse through the Tor network.

It can also access **.onion addresses**.

Because Apple requires iOS browsers to use certain Apple browser technology, Onion Browser cannot provide all of the same protections available through the official Tor Browser on computers and Android.

Good for: People using iPhones or iPads who need access to Tor.

DUCKDUCKGO BROWSER — FREE

Website: duckduckgo.com

DuckDuckGo provides a privacy-focused browser designed primarily to reduce everyday tracking.

It can block many third-party trackers, tracking cookies, and other common tracking technologies.

It is available for:

- Windows
- Mac
- Android
- iPhone
- iPad

IMPORTANT:

DuckDuckGo Browser is NOT Tor.

The free DuckDuckGo Browser is NOT a VPN.

DuckDuckGo does offer a VPN through its paid Privacy Pro subscription, but that is separate from its free browser.

Good for: Everyday browsing when someone wants substantially less advertising and behavioral tracking without changing how they normally use the internet.

PROTON VPN FREE — FREE

Website: protonvpn.com

A VPN — Virtual Private Network — creates an encrypted connection between your device and the VPN company's server.

Instead of websites generally seeing your normal home/public IP address, they see the IP address of the VPN server.

A VPN can also prevent the local network or internet provider from directly seeing the destinations you connect to in the same way as an ordinary unprotected connection.

Proton VPN offers a legitimate free tier with unlimited data and no advertising.

Available for:

- Windows
- Mac
- Linux
- Android
- iPhone/iPad

IMPORTANT:

VPN ≠ complete anonymity.

Using a VPN means placing significant trust in the VPN provider. This is why downloading random "FREE VPN" applications is not recommended. Some free VPN providers may make money through advertising, tracking, or collection of user data.

Good for: Protecting network traffic, public Wi-Fi use, concealing your normal IP address from websites, and general internet privacy.

MULLVAD BROWSER — FREE

Website: mullvad.net/browser

Mullvad Browser was developed in collaboration with the Tor Project.

It incorporates many technologies designed to resist tracking and browser fingerprinting.

However:

Mullvad Browser DOES NOT automatically connect through Tor.

Think of it this way:

Tor Browser = Privacy-hardened browser + Tor network

Mullvad Browser = Privacy-hardened browser without the Tor network

Mullvad Browser can instead be used alongside a trusted VPN.

Good for: People who want strong browser-level privacy and fingerprinting protection without routing their browsing through Tor.

WHAT IS A .ONION WEBSITE?

“.onion” is not another browser.

It is a special type of internet address used by **Tor Onion Services**.

A website or organization may operate a regular website while also maintaining a separate .onion address accessible through Tor.

Onion services have legitimate uses including:

- Privacy-preserving communication
- Journalism
- Whistleblower submission systems
- Censorship resistance
- Protecting the physical location of servers
- Protecting users' originating locations

The phrase “**dark web**” is frequently associated with .onion websites, but Tor itself is simply privacy and anonymity technology. As with the regular internet, what matters is how someone uses it.

WHICH ONE SHOULD I USE?

I JUST WANT LESS TRACKING

Use:

DuckDuckGo Browser

Simple, free, and available on most devices.

I WANT TO PROTECT MY DEVICE'S INTERNET CONNECTION

Use:

Proton VPN Free

This protects network traffic beyond a single browser.

I WANT STRONGER ANONYMOUS BROWSING

Use:

Tor Browser

For Windows, Mac, Linux, and Android.

For iPhone/iPad:

Onion Browser

I WANT STRONG BROWSER PRIVACY WITHOUT USING TOR

Use:

Mullvad Browser

It can also be combined with a reputable VPN.

WHAT THESE TOOLS DO NOT DO

VPNs, Tor, and private browsers **do not magically make someone invisible online.**

For example, if someone opens Tor Browser and then signs into their regular Facebook, Gmail, TikTok, or another personally identifiable account, that service still knows which account is being used.

A VPN does not automatically protect against malware.

A VPN does not make someone anonymous to a website they voluntarily identify themselves to.

Private/Incognito Mode does not make someone anonymous online. It primarily prevents certain browsing information from being saved locally on that device.

Tor does not protect information that someone voluntarily reveals.

Installing random extensions in Tor Browser can also undermine its privacy protections and potentially make the browser easier to identify through fingerprinting.

More privacy applications do not automatically equal more privacy. An unusual or incorrectly configured setup can sometimes make a device or browser more identifiable.

SIMPLE FREE PRIVACY TOOLKIT

EVERYDAY PRIVATE BROWSING

DuckDuckGo Browser

DEVICE-WIDE VPN

Proton VPN Free

TOR / STRONGER ANONYMOUS BROWSING

Tor Browser

TOR ACCESS ON IPHONE/IPAD

Onion Browser

ANTI-TRACKING / ANTI-FINGERPRINTING BROWSER

Mullvad Browser

The best option depends on the goal. Someone trying to reduce advertising trackers has very different needs from someone who needs to conceal their originating IP address or access the Tor network.